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Course: 21CSC202J - Operating Systems

Unit – 2 : Session – 15 : SLO – 1 | Dining Philosophers — Missing Pseudocode

Answers

TODO 1: wait(fork[i])

TODO 2: wait(fork[(i + 1) % 5])

TODO 3: signal(fork[i])

TODO 4: signal(fork[(i + 1) % 5])

Deadlock-free version

```
semaphore fork[5] = {1, 1, 1, 1, 1}

function philosopher(i):
    while true:
        think()
        if i == 4:                    // last philosopher reverses order
            wait(fork[(i + 1) % 5])
            wait(fork[i])
        else:
            wait(fork[i])
            wait(fork[(i + 1) % 5])
        eat()
        signal(fork[i])
        signal(fork[(i + 1) % 5])
```

Why the ordering matters

1. If every philosopher takes the left fork first, all five can hold one fork and wait
2. for the next -> circular wait -> deadlock. Reversing the order for one philosopher
3. breaks the cycle, so at least one philosopher always gets both forks.